

Jaspreet Singh Nahal

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Final-year Computer Science student specializing in AI/ML Engineering and Full Stack Development. Proficient in Python, PyTorch, LangChain, Hugging Face Transformers, and React - with hands-on experience building LLM pipelines, RAG systems, and FastAPI services. Pretrained a 205M-parameter GPT Transformer from scratch. Two-time competition winner.

EXPERIENCE

Machine Learning Intern

Aug 2025 – Oct 2025

Expedien eSolutions

Noida, India

- Engineered a RAG-based legal query resolver using LangChain, FAISS, and Hugging Face Transformers, enabling accurate document-grounded responses over a corpus of 10,000+ legal documents
- Designed and optimized the document retrieval pipeline using vector embeddings and semantic search, achieving sub-2 second end-to-end response time with a Streamlit UI, improving query throughput by 40%
- Applied instruction fine-tuning techniques and prompt engineering to improve response quality and reduce hallucinations across the production pipeline

Algorithmic Trading Developer

Nov 2024 – Dec 2024

Freelance

Remote

- Developed an automated stock trading system using the SMA Crossover strategy with a Flask REST API and PostgreSQL database, delivering consistent simulated returns during backtesting
- Reduced trade execution latency by 30% through async processing, optimized SQL queries, and efficient data pipeline architecture

PROJECTS

EverydayGPT | [Live Demo](#) | *PyTorch, Transformers, FAISS, RAG, NLP*

Feb 2026

- Pretrained a 205M-parameter GPT-style Transformer from scratch in PyTorch using causal self-attention and GPT-2 tokenization on a subset of 10 billion token FineWeb-Edu and Wikipedia corpus
- Designed a Retrieval-Augmented Generation pipeline using SentenceTransformers and FAISS achieving sub-205ms retrieval latency; reduced hallucinations via confidence-based gating with instruction fine-tuning, top-k sampling, and repetition penalty for stable multi-turn performance

Aria | *LangGraph, FastAPI, Ollama, RAG, FAISS, React, Python* | *In Progress*

May 2026

- Building a local-first autonomous desktop employee that converts natural language instructions into structured OS-level task execution across files, email, and browser with zero cloud dependency
- Architecting a multi-agent debate layer using LangGraph where competing sub-agents verify action correctness before irreversible execution; implementing tiered safety model classifying actions as auto-execute, confirm, or reject
- Designing persistent memory system storing organizational context, user preferences, and task history enabling cross-session awareness; stack includes FastAPI backend, Ollama local inference, FAISS retrieval, and React frontend

CodeAtlas | *Python, FastAPI, LangChain, RAG, NetworkX, React, TypeScript*

Mar 2026

- Architected an AI-powered code intelligence platform with a RAG pipeline over GitHub repositories and ZIP archives, enabling natural language Q&A, semantic concept extraction, and dependency graph generation across 8 programming languages
- Implemented domain-aware file classification and an adaptive learning layer personalizing code explanations by proficiency; deployed via async FastAPI with a React force-graph UI for real-time codebase exploration

EDUCATION

Bachelor of Technology in Computer Science Engineering

2023 – 2027

Dr. A.P.J. Abdul Kalam Technical University

Uttar Pradesh, India

SKILLS

Programming Languages: Python, Java, C++, JavaScript, TypeScript

Machine Learning and AI: PyTorch, Hugging Face Transformers, SentenceTransformers, FAISS, LangChain, Retrieval-Augmented Generation (RAG), NLP, Instruction Fine-Tuning, large-scale pretraining

Data Science: NumPy, Pandas, scikit-learn, Spark, NetworkX, gradient-based optimization

Backend and DevOps: FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, Git, Docker, GitHub Actions, Streamlit

Frontend: React.js, TypeScript, D3.js, Vite, TailwindCSS

CERTIFICATIONS

Google: Prompt Design in Vertex AI, Responsible AI Practices

IBM: Accelerating Deep Learning with GPUs

BCG X via Forage: GenAI Financial Chatbot Job Simulation

Competitions: 1st Place, Code Farming, 643 participants (2026); 1st Place, DSA Championship 24-hour contest (2026); 3rd Place, HackIndia Spark-4 Voyager Blitz (Mar 2026)

Hackathons: IBM AI Hackathon, Google Solutions Challenge, GUVI Hackathon (2025)